Name	Junjie Huang
Student ID	33048266
Studio Session	16

Part A: URL of the Visualisation

Paste your visualisation URL in the empty space below.
Tip: Test your URL in incognito mode to ensure it is a working URL
https://guiltyeaten.github.io/FIT3179DataVisualization2/

Part B: Domain Description

Describe your domain chosen for DV II

Topic: Graduate Starting Salaries
Investigate graduate salaries across Australian fields of study. Compare starting salaries by discipline, gender or employment status. Explore how graduate earnings have changed over time.

The domain of this data visualization is to investigate graduate salaries across Australian fields of study. It focuses on critically evaluating the return on investment of university degrees by analysing graduate starting salaries, full-time employment rates, and job-match quality across different fields of study, as well as analysing the impact of universities, geographic regions and demographics on graduate salaries.

Describe the Why and Who aspects of your DV II

Why:
Choosing a university degree is one of the most important decisions for young Australians to make. The purpose of this visualisation is to provide a transparent, data-driven analysis of graduate outcomes. It aims to answer key questions: Which fields offer the best financial return? Does attending a prestigious "Group of Eight" university guarantee a higher salary? What are the hidden trade-offs between earning a high salary and securing a job? The goal is to reveal the truth of higher education with actual data, which gives the young Australian the full picture of this important decision they need to make to allow them to make the best decision for themselves.

Who:

This visualization is designed for the average Australian, specifically prospective university students, current students, and their parents, who need accessible information to make informed decisions about their education. It is also valuable for career counsellors and policymakers evaluating the alignment between higher education offerings and labour market demands. The visualisation is easily understandable to a general audience by avoiding complex statistical tools and methods.

Describe the "What" aspect of your DV II

The data visualization investigates the topic of Graduate Starting Salaries from several directions, combining two datasets:

Graduate Outcomes (QILT GOS): Data from the Quality Indicators for Learning and Teaching (QILT) Graduate Outcomes Survey (2020–2025). This includes median starting salaries, full-time employment rates, overqualification rates, and international/domestic salary splits across 21 broad fields of study and various Australian universities.

General Economic Context (ABS): Data from the Australian Bureau of Statistics (ABS) Personal Income in Australia (2022–23). This includes median personal income and the SA4 geographic regions data.

Describe the "How" aspect of your DV II

Chart 1: Return on Education

Source: QILT GOS

Idiom: Lollipop Chart with Range Bars

Why:
A lollipop chart shows the average salary to a $0 baseline does the same job as the normal bar chart, but the dot at the end emphasises the average salary for that specific education level. It also uses grey range bars to clearly show the massive salary variance within each qualification level between different fields.

How:
Dot marks for the average salary, a grey bar to the $0 axis, and a thick grey band (range bar) showing the min-max spread of the 21 fields for the same education level.

Chart 2: Regional Income Baseline

Source: ABS

Idiom: Choropleth Map

Why:
Shows the average income level in general for each region based on SA4 geo data. This describes the economic level within the use Australia map, provides the background knowledge and tries to make a connection with the next chart to tell the story and answer the question that is the university in the wealthy area always providing a higher starting salary?

How:
Geoshape marks for SA4 regions. Colour brightness as the channel for median income, provides an interactive dropdown to zoom into major capital cities to ensure good readability for crowded areas in major cities.

Chart 3: University Salaries

Source: QILT GOS

Idiom: Colour dot Map

Why:
To challenge the myth that metro universities pay the best. Plotting dots on a map allows the reader to see that a regional university (like USQ) can out-earn metropolitan ones, which busts the myth. It also allows the reader to gain more information by checking the starting salary in each university and comparing them for each area.

How:
Dot marks positioned using longitude/latitude, which these data collect from OSM. Colour brightness as the channel for median starting salary. Use the same SA4 data as previous map.

Chart 4: University Employability

Source: QILT GOS

Idiom: Proportional Symbol Map

Why:
To contrast "salary" with "employability." It visually proves that prestigious Group of Eight universities don't always have the highest full-time employment rates; regional universities often have larger graduate pipelines to local employers. This provides a different view than

the previous map, as it introduces employability as another important factor when picking a university. If a university has extremely high starting salary but a low employment rate, readers need to choose carefully. Because if you are not employed, you are getting 0 salary that's not showing in the record.

How:
Dot marks, size-area as the channel for full-time employment rate. Same SA4 geo data used as previous 2 maps.

Chart 5: Postgrad vs Undergrad Spread

Source: QILT GOS

Idiom: Jitter Dot Plot (Strip Plot)

Why:
Reveals that undergraduate salaries are tightly clustered across all universities, while postgraduate salaries have a massive spread. A jitter plot prevents overplotting and shows the density of universities at different salary levels. It shows that for undergraduate salaries, the gap is smaller between different universities, and the gap gets much bigger when it comes to postgraduate salaries. The 2 clusters show how much difference postgraduate salaries can make, and the gap tells the importance of picking the right university for your postgraduate degree.

How:
Dot marks, x-axis for salary, y-axis for categories for education level, which are undergraduates and postgraduates.

Chart 6: Salary Growth (2020-2025)

Source: QILT GOS

Idiom: Slope Chart

Why: To track the trajectory of all 21 fields over five years. It highlights that while all fields have their starting salaries grow within 5 years, the gap is narrowing except Dentistry.

How: Line marks connecting the 2020 salary to the 2025 salary, colour hue for different fields of study.

Chart 7: Field Ranking Stability

Source: QILT GOS

Idiom: Heatmap

Why: connects with previous chart, provides more detail for how much salary each field grows every year and shows the parallel growth pattern. Heatmap is the best idiom to identify clusters and hierarchies across two dimensions (field and year).

How: area marks, x-axis for year, y-axis for field of study (sorted by salary), colour brightness for salary.

Chart 8: The Salary/Employment Triangle

Source: QILT GOS

Idiom: Ternary Plot

Why:
Used to show the balance between three competing outcomes: Salary, Full-Time Employment, and Job-Match (low overqualification). It shows that some fields are more balanced than others, while some fields of study lean more towards 1 or 2 outcomes but have significantly weaker outcomes. It helps to clarify requirements and helps students to make decisions, as picking a certain field of study can be in a bad position in salary but gains advantages in other places.

How:
Custom calculated X/Y coordinates to map the three weights inside a triangle. Dot marks, x-y position as the channel for balance.

Chart 9: Rank Trade-offs

Source: QILT GOS

Idiom: Bump Chart

Why:
To show the brutal trade-offs hidden in the ternary plot. It reveals that a field in the middle of the triangle is not necessarily perfect, as they having all 3 ranks at the bottom or all 3 ranks at the top of the rank. This chart provides the extra information that the ternary chart cannot show, and provides more information to the readers about the advantages and disadvantages of each field of study.

How:
Line marks connecting the ranks, with circle marks and text labels containing the rank number, colour hue for fields of study

Chart 10: Salary vs Employment over Time

Source: QILT GOS

Idiom: Interactive Scatter Plot

Why:
Allows the user to drag a slider and physically watch how the relationship between salary and employment moves year by year for fields of study.

How:
Dot marks, x-axis for employment rate, y-axis for salary. A year slider is provided for the reader to change the years of data presented.

Chart 11: International vs Domestic Gap

Source: QILT GOS

Idiom: Multi-line Chart with Gap Area

Why: Shows the narrowing gap between international and domestic graduate salaries. The shaded area between the lines makes the magnitude of the gap immediately visible.

How: Line marks for the two category groups, with a shaded area mark (coloured grey) filling the space between them to show the difference

Chart 12: The Gender Pay Gap

Source: QILT GOS

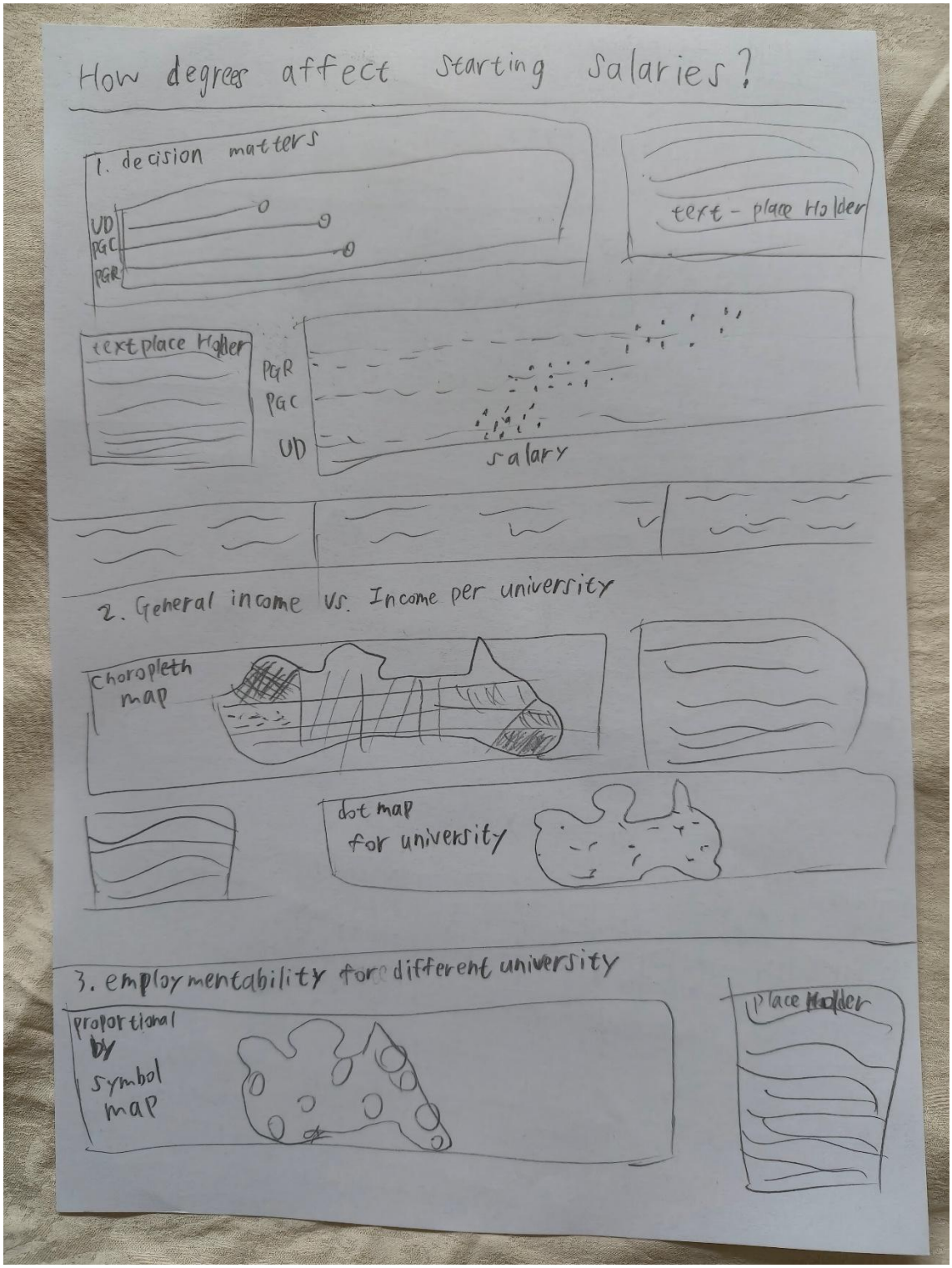
Idiom: Dumbbell Chart

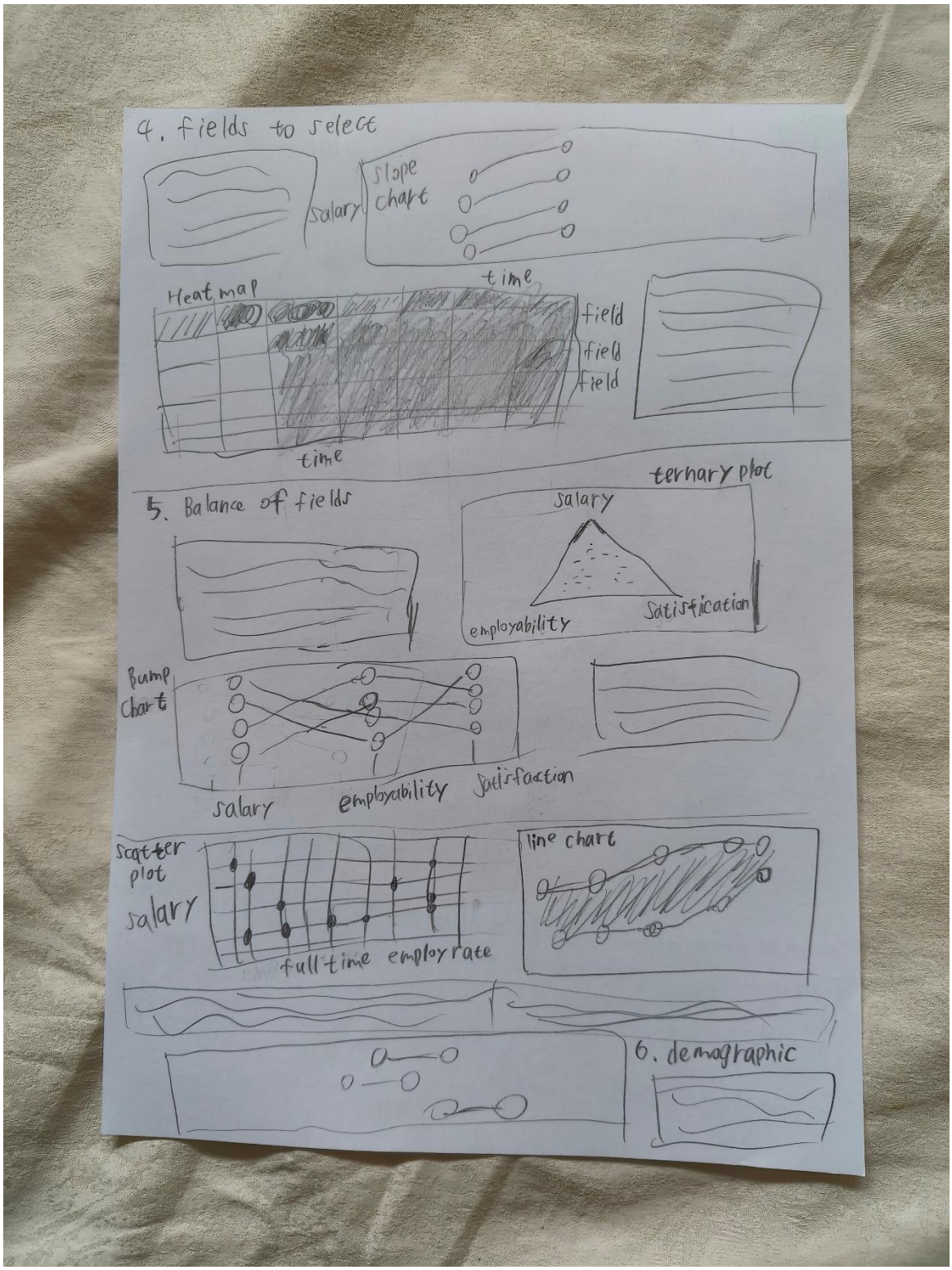
Why: A dumbbell chart is highly effective for showing the gap between two related data points (Male vs. Female) across different fields of study. It highlights that some fields might be in favour of females and some of the field is the opposite.

How: Line marks connect the male dot to the female dot for each field. Colour hue (light blue/orange) as the channel for the genders.

Part C: Sketch

A URL to a PDF document with an image of your sketch on your GitHub account.
Tip: Test your URL in incognito mode to ensure it is a working URL



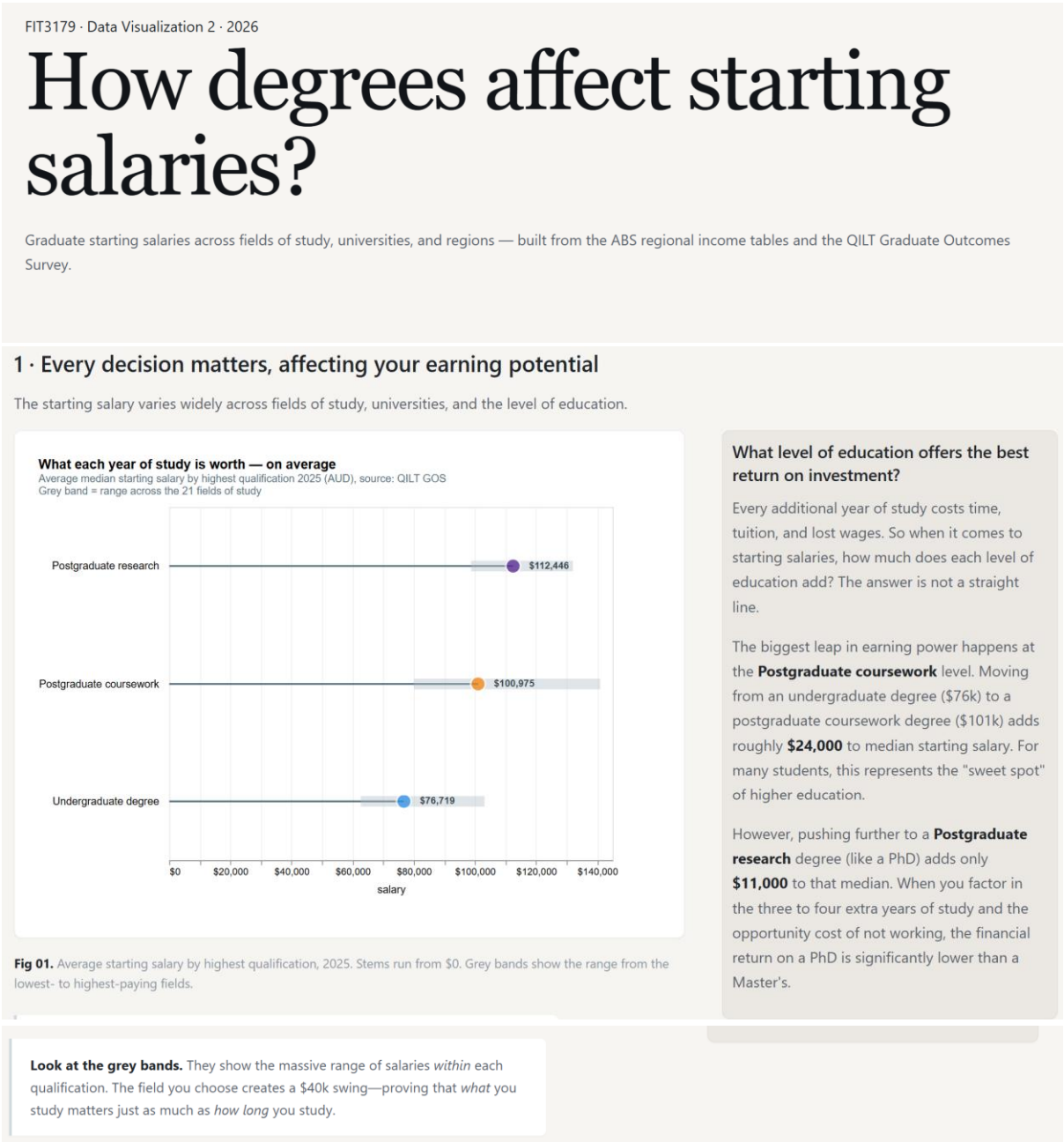


Paste a screenshot of your sketch for your DV II.

Note: Use the entire page here for the screenshot and ensure that it is clearly legible and visible

Part D: Full-size screenshot of your DV II

Paste a full-size screenshot of your DV II assignment from the URL here



What gives the best return on investment in higher education?

The answer is postgraduate. Undergraduate salaries cluster tightly between **$67k and $87k** across almost every Australian university. It doesn't matter much which one you attend for your bachelor's degree; the field you study is the dominant factor.

But at the postgraduate level, the spread is nearly twice as wide (**$90k to $135k**). A postgraduate degree from a top-tier institution—or one with strong industry connections—can push your starting salary far above the national average. If you are going to pay for a Master's, the university you choose matters significantly more.

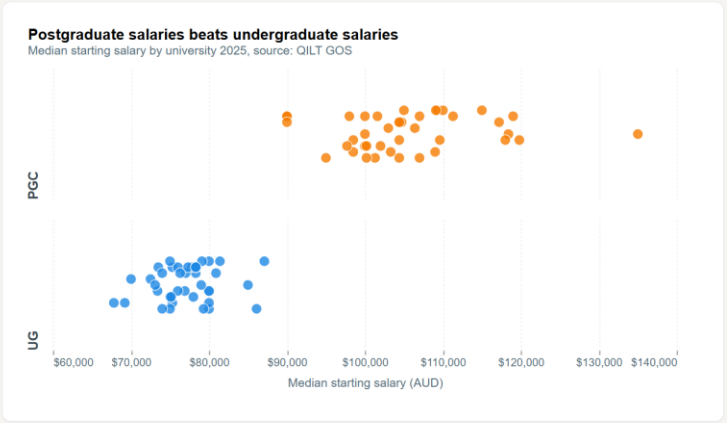


Fig 05. Jitter dot plot. Each dot is a university. Horizontal position = median starting salary. Vertical jitter is for readability only.

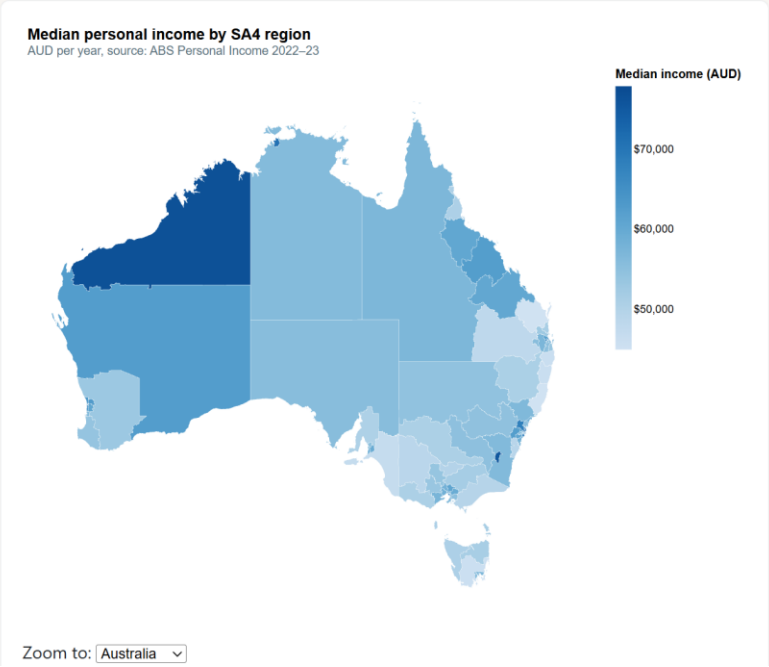
$76,719
Average undergraduate starting salary across 21 broad fields (2025)

$100,975
Average postgraduate coursework starting salary

~$20k
Spread between the highest- and lowest-paying undergraduate fields on different universities

2 · The regional income level is uneven over Australia, but income level does not determine graduate starting salaries

Australia's highest earners cluster in a handful of inner-city regions; the lowest cluster in remote and regional areas. Two maps establish the economic context for everything that follows.



Where are the highest earners located?

Before looking at graduates, we need to establish the economic baseline they are entering. The darkest regions—Sydney's North Shore, the WA Outback North, and the ACT—are the highest earners in the country. Conversely, regions like Wide Bay in Queensland, South-East Tasmania, and the Mid North Coast in NSW represent the lowest income levels.

This chart shows the average income level for everyone in a region, not just graduate students. But it establishes the economic context for the job market that graduates are entering.

Average income only tells half the story. For a university student, the question is: **How much *can they get* for their starting salary?** Do graduates follow this exact same geographic divide, or does the specific university they attend matter more?

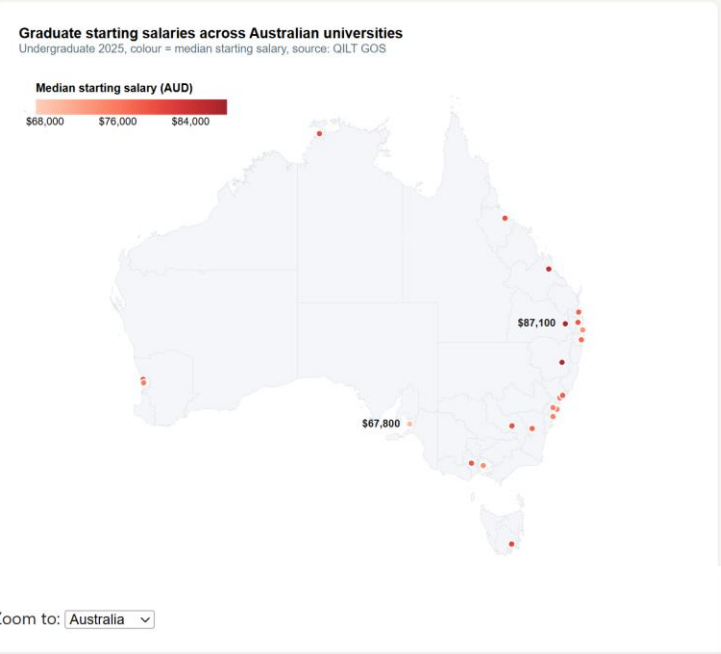
Fig 02. Median personal income by SA4 region, 2022–23 (ABS). Darker blue represents higher income.

The next chart explores the median starting salaries of graduates from different universities across Australia.

Do Melbourne and Sydney always pay the most?

This map plots every Australian university recorded on QILT at its real geographic location. The colour of each dot represents the median starting salary for its 2025 undergraduate cohort, from the palest pink (lowest salaries) to the darkest red (highest salaries).

The common assumption is that studying in a major city—like Sydney or Melbourne—guarantees the highest starting salary. But the data busts that myth. The highest-paying university in the country (the darkest red dot at **$87,100**) is the University of Southern Queensland, located in the regional town of Toowoomba. Meanwhile, some universities in major metro areas (like the lightest dot in Adelaide at **$67,800**) sit at the bottom of the pack.



But salary is only one half of the equation. A high-paying job is useless if you can't get one. The next map asks: if we look at *employment rates* instead of salaries, do these same universities still come out on top?

Fig 03. Median starting salary by university, 2025 (QILT GOS). Darker red indicates a higher median starting salary. Use the dropdown to zoom into metropolitan clusters.

3 · Employability tells a different story for different universities

The common assumption is that popular, big-city universities produce the highest-paid graduates with higher employability. The data suggests otherwise.



Do the most prestigious universities offer better employment outcomes?

This map plots every Australian university at its geographic location. The size of each bubble represents its full-time employment rate—the percentage of 2025 undergraduates who found a full-time job within four months of finishing.

The common assumption is that the prestigious Group of Eight (in the major capital cities) guarantee the best employment outcomes. But the data busts that myth. Some of the **largest bubbles** on the map belong to regional universities, such as Charles Sturt University and James Cook University.

Why do regional universities perform so well here? These institutions often have incredibly strong ties to local health, education, and agriculture employers, creating a direct pipeline from graduation to a full-time job. Prestige doesn't always equal employability.

Fig 04. Proportional symbol map of full-time employment rates by university. QILT GOS 2025.

4 · The field you choose matters more than the university you attend

Salaries have risen roughly 30% across every field since 2020, and the ranking of those fields has barely moved. **Dentistry** leads. **Pharmacy** trails.

How has the salary landscape shifted over five years?

This slope chart tracks every single undergraduate field of study between 2020 and 2025. At first glance, the overwhelming pattern is upward movement. Every field experienced a salary increase over this period, generally rising by roughly 30%.

Notice how the lines run almost perfectly parallel. This is the crucial insight: **while the whole economy inflated, the relative hierarchy of these fields never changed.** A field that was at the top in 2020 is still at the top today, and those at the bottom stayed at the bottom.

Over 5 years, **Dentistry** started on $84,000 and jumped to **$103,100**. Meanwhile, **Pharmacy** started on just $49,600 and only reached **$62,600**. While **Dentistry** leads the pack and makes the salary gap greater, the rest of the fields are more clustered together over the 5 years.

The spread between the highest and lowest earning fields is a massive **$40,000**, and that gap has not narrowed in five years. Your choice of field locks in your baseline salary.

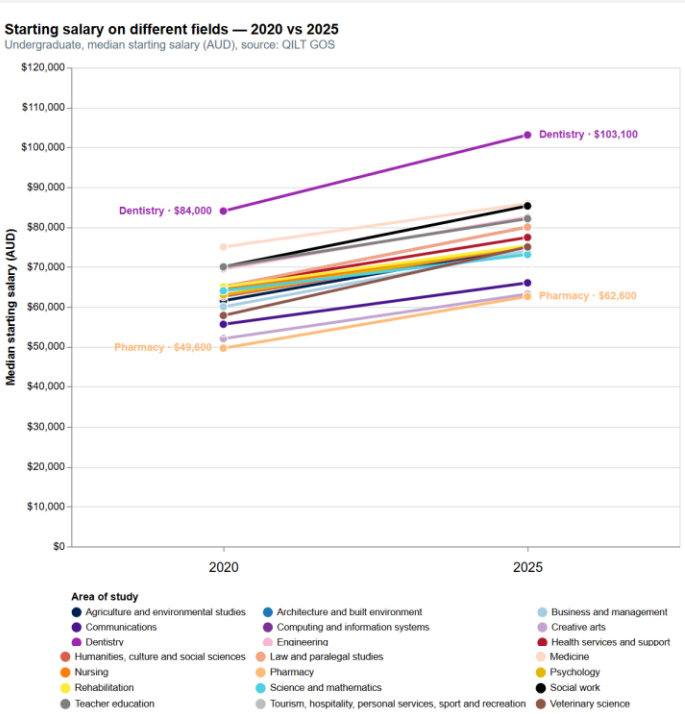
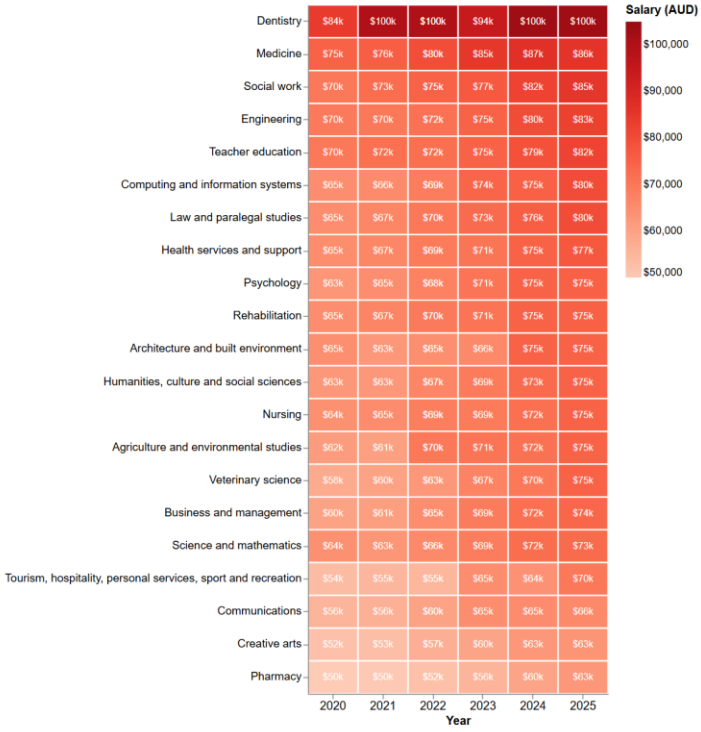


Fig 06. Slope chart. Median starting salary for all 21 undergraduate fields in 2020 vs 2025.



Is the ranking of fields stable over time?

The lines in the slope chart shows a parallel growth, but this heatmap proves just how deeply entrenched those rankings are. Here, every field is tracked across six years. Darker red cells represent higher salaries, and the fields are sorted from highest-paying at the top to lowest at the bottom.

If the graduate job market were volatile, you would see the dark bands zig-zagging up and down as fields traded places. Instead, look at the horizontal bands—they barely shift.

Even through a global pandemic, a cost-of-living crisis, and massive economic shifts, the vertical hierarchy remained practically frozen. No field managed to jump more than three places in the salary rankings.

This tells us that the graduate job market is highly segmented. Your choice of field essentially locks in your starting salary bracket. Climbing from the bottom rows (like **Pharmacy** or **Creative Arts**) to the top rows (**Dentistry** or **Medicine**) is incredibly difficult, even over half a decade.

But salary isn't everything, there are other things you should consider when choosing a



Fig 07. Heatmap of median undergraduate salary by field and year, 2020–2025. Fields sorted by 2025 salary.

But salary isn't everything, there are other things you should consider when choosing a field, such as employment and job-match. The next section explores the trade-offs between these three outcomes.

5 · No field wins on everything at once

Which fields look balanced across multiple outcomes, and which fields lean heavily in one direction?

Can one field dominate in salary, employment, and job-match simultaneously?

Each dot is a field. Position shows where its profile leans—toward salary strength, employment strength, or job-match strength. A dot near the centre is equally strong (or weak) on all three.

Look at the extremes. **Pharmacy** sits at the bottom edge: it has one of the strongest employment profiles, but the weakest salary. **Creative Arts** leans hard toward the job-match corner, reflecting the fact that graduates feel they are using their degree, even though both salary and employment strength are low.

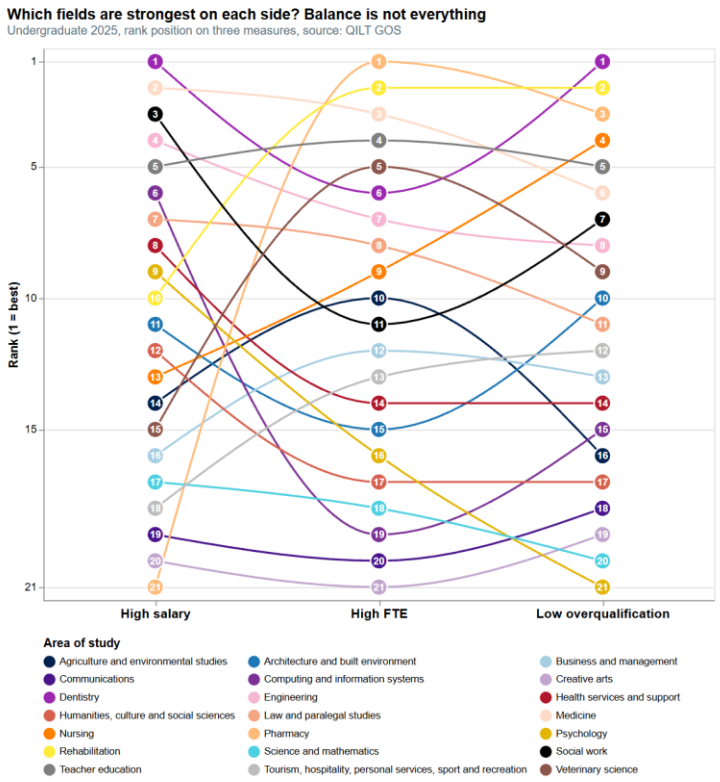
A field like **Medicine** sits towards the top: strong salary and strong job-match. But it's important to note what this triangle *doesn't* show—the human cost. The fields at the top of the triangle (**Medicine**, **Dentistry**) require years of additional study and significant financial investment.

The centre of the triangle is often the most deceptive. A dot in the "balanced zone" isn't necessarily winning; it often just means the field is average at everything. It lacks a standout reason to choose it.

But this chart only shows the *shape* of the profile; it doesn't tell us how extreme the differences are. To see the magnitude of these trade-offs, we have to look at the ranks.



Fig 08. Ternary balance plot. Undergraduate 2025.



Does a "balanced" profile mean a field is actually a winner?

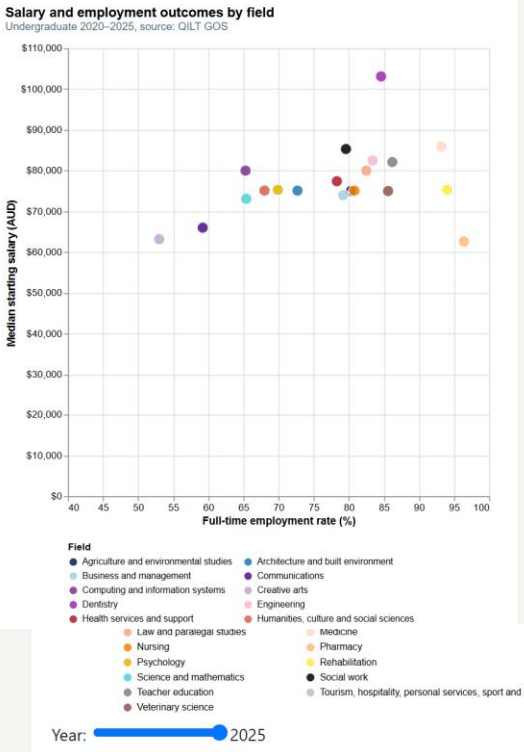
The ternary plot above shows the *shape* of a field's profile. A dot near the centre might look "balanced," but it can also mean a field is simply average at everything. To see the real stakes, we have to look at the **ranks**.

This bump chart reveals the brutal trade-offs hiding behind the profiles. **Pharmacy** sits mid-pack in the triangle, but here you can see it is actually **Rank 1 for employment** and **Rank 21 for salary**. Conversely, **Dentistry** is Rank 1 for salary but only mid-tier for employment.

The rank on each axis tells the real story of what you are trading away when you choose a field. The centre of the triangle is a trap: it hides the fact that you might be sacrificing top-tier earnings for top-tier employability.

> You cannot have it all. The field that gets you the highest salary is rarely the same field that gets you the highest employment rate.

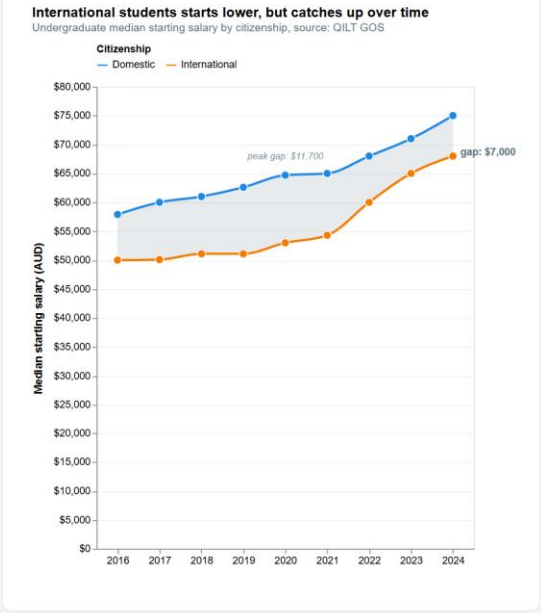
Fig 09. Rank position of each field across salary, full-time employment, and job-match (low overqualification), 2025. Coloured lines highlight extreme trade-offs.



Do higher salaries come with better employment?

Rarely. Drag the slider to watch the years tick by. The dots move, but the relationship stays the same: fields like **Pharmacy** and **Rehabilitation** sit at the top of the employment axis but the bottom of the salary axis. High pay and high employment are often a trade-off, not a package deal.

Fig 10. Animated scatter of salary vs full-time employment, 2020–2025.



Does citizenship status affect your starting salary?

Yes, but the gap is narrowing. The international–domestic salary gap widened sharply through 2022, but has been closing since. Both groups' salaries grew steadily, but the recent narrowing likely reflects a shift in the fields international students are choosing, rather than a sudden change in pay equity.

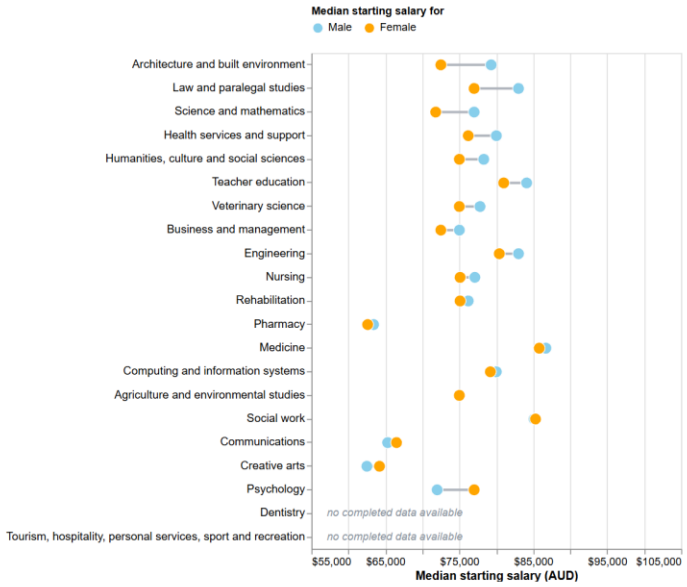
Fig 11. Median starting salary by citizenship, 2016–2024.

6 · Beyond the headline number: gender and qualification level

Treating all graduates in a field as identical hides two important variations. Gender changes the picture, and so does how far you take your education.

Gender pay gap by field of study

Undergraduate 2025, sorted by size of gap from male to female, source: QILT GOS



Is the gender pay gap universal across all fields?

Female graduates earn less than male graduates in almost every single field, even at the very first step of their careers. But the size of the gap depends heavily on the discipline.

The gap is widest in **Architecture** and **Law and paralegal studies** (around 8–10%), and smallest in **Agriculture and environmental studies** and **Social work**, where the workforce is highly feminised and award wages dominate.

The story isn't one of universal disparity—it's that the gap is highly context-dependent. This is crucial for students interpreting the "typical salary" for their chosen discipline. The median salary advertised for a field might hide a significant divergence between what men and women are actually offered.

Fig 12. Male vs female median starting salary, by broad field. Undergraduate 2025. Sorted by the size of the gap.

Three things a student can take from this data

1 · Interest first

There is no perfect field to study, fields that are strong on some outcome and weak on others. There is always a trade-off. The difficulty of the field is not shown in this study, fields like **Dentistry** with higher rank on every outcome comes with a much higher workload and longer study time. Students should prioritise interest first, and treat salary and employment as secondary considerations.

2 · Selection of University is important

The choice of university matters in employment and salary outcomes, especially at postgraduate level. Some universities have strong ties to local employers which have more employment opportunities, while some universities have higher starting salaries. Students should consider the university's location, reputation, and graduate outcomes when making their decision.

3 · Master's degrees offer the best return

Master's degrees are the only postgraduate qualification that consistently out-earn undergraduate degrees in terms of starting salaries. PhDs are the least rewarded in salary, despite the fact that it requires significant more time and financial investment.

Sources and notes

Data Sources

- **Graduate Outcomes:** QILT Graduate Outcomes Survey (2016–2025)
- **Regional Income & Inequality:** ABS Personal Income in Australia (2022–23)
- **Geographic Boundaries:** ABS Australian Statistical Geography Standard (ASGS 2021)
- **University Coordinates:** OpenStreetMap (Compiled for primary campuses)

Author Information

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Student ID: 33048266
FIT3179 · Data Visualisation 2

AI Acknowledgement

Generative AI was used to assist with Vega-Lite specification syntax, CSS layout scaffolding, and grammar-checking of narrative text.